Huawei Access Node KOR-ACC-83 Hardware/Environmental Fault Resolution Procedure

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6293

Description

The occurrence of hardware/environmental faults, such as ASIC memory parity errors, line card crashes, and power supply redundancy loss, in the Huawei access node KOR-ACC-83 situated at the Koramangala Datacenter, necessitates a comprehensive understanding of the underlying architectural components and protocol mechanics. The Huawei access node, operating as a critical aggregation point for subscriber traffic, relies heavily on the integrity of its line cards, which house the ASICs responsible for packet processing and forwarding. Memory parity errors within these ASICs can lead to unpredictable system behavior, including packet corruption and line card crashes, thereby impacting service availability and quality. Furthermore, the loss of power supply redundancy can exacerbate the situation by introducing single points of failure, potentially leading to node-wide outages. The interplay between these hardware components and the environmental conditions (such as temperature and humidity) at the Koramangala Datacenter can also contribute to the manifestation of such faults. Thus, a methodical approach to diagnosing and resolving these issues, adhering strictly to Huawei's CLI dialect, is paramount to ensure prompt restoration of services and prevention of future occurrences.

Pre-requisite Checks

- Dispatch remote hands to Koramangala Datacenter for physical inspection and intervention if necessary.

- Verify the status of redundant links to ensure no single points of failure exist in the network topology.

- Check environmental conditions (temperature, humidity) at the datacenter to ensure they are within the recommended ranges for the Huawei equipment.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware degradation or manufacturing defects.	High
Line card crashes resulting from software bugs or incompatible firmware versions.	Medium
Power supply redundancy loss due to faulty PSU or improper installation.	Low

Diagnostic Commands & Expected Outputs

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Command: display device, Expected Output: Device information, including model, version, and current status., Description: This command isolates issues related to device hardware and firmware versions, providing a baseline for further troubleshooting.
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```
Command: display alarm, Expected Output: Current alarm list, including severity and description., Description: This command helps in identifying the specific alarm codes (like ALARM_6293) and their implications on the system's operational state.

Command: display environment, Expected Output: Environmental parameters such as temperature, voltage, and fan status., Description: This command is crucial for assessing the environmental conditions that could be contributing to the hardware faults, ensuring they are within the specified operating ranges.
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Resolution Steps

Step 1: Step: Replace Faulty Line Card, Command: replace board , Description: Physically replacing the line card identified as faulty, ensuring that the replacement is compatible and properly seated, and then using the Huawei CLI to recognize and configure the new board.

Step 2: Step: Update Firmware, Command: boot-loader update , Description: Updating the firmware of the affected line card or the entire system to the latest version, ensuring compatibility and resolving any known bugs that could lead to hardware faults.

Step 3: Step: Restore Power Supply Redundancy, Command: power supply backup enable, Description: Enabling or replacing the power supply unit to restore redundancy, ensuring that the system has a backup power source in case of a primary power supply failure.

Post-Check Verification

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Command: display device detail, Expected Output: Detailed device information, including the status of all components., Description: Verifies that all hardware components, including the newly replaced or updated parts, are recognized and functioning correctly by the system.

Command: display alarm all, Expected Output: List of all current alarms, which should not include ALARM_6293 after resolution., Description: Confirms that the specific alarm related to the hardware/environmental fault has been cleared, indicating successful resolution of the issue.
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Roll-back Procedures

- Step: Revert to Previous Firmware, Command: boot-loader revert , Description: If the firmware update causes unforeseen compatibility issues, reverting to the previous firmware version to restore system stability.

- Step: Remove and Reinsert Line Card, Command: remove board then insert board , Description: If the replacement line card does not resolve the issue, removing and then reinserting it to ensure proper seating and system recognition.